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On the MgF_2–YF_3 System

S.N. Achary, S.J. Patwe, A.K. Tyagi*

Applied Chemistry Division, Bhabha Atomic Research Centre, Trombay, Mumbai 400085, India

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Abstract

Earlier we reported the room-temperature phase equilibria in MF_2–YF_3 (M = Ba, Sr, and Ca) systems. In the present study, synthesis and characterization of the $Y_{1-x}Mg_{2+1.5x}F_7$ ($-1.33 \leq x \leq 1.0$) compounds were attempted, to establish a similar phase relation in the MgF_2–YF_3 system. Neither YF_3 in MgF_2 nor MgF_2 in YF_3 was found to have solubility under slow-cooled conditions. © 2000 Elsevier Science Ltd. All rights reserved.

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1. Introduction

In the quest of stable phases in the binary fluoride systems, we have been investigating the phase relations in the MF_2–YF_3 systems. A large number of compounds with general composition $Y_{1-x}M_{2+1.5x}F_7$ where M = Ca^{2+}, Sr^{2+}, and Ba^{2+} were prepared and well characterized [1–3]. It was found that the nature of heat-treatment has a significant influence in the formation of the products and solubility limits. We reported that the solubility limit of YF_3 in MF_2 lattice decreases with increase in the size difference of Y^{3+} and M^{2+} ions. In the SrF_2–YF_3 and CaF_2–YF_3 systems, a tysonite-type single-phase compound with composition at about 75 mol% of YF_3 was observed. The cubic YCa_2F_7 represents the upper solubility limit of YF_3 in CaF_2, whereas YSr_2F_7 has a tetragonal distortion. However, YBa_2F_7 does not exist as a single phase; it is a mixture of a rhombohedral-type and cubic fluorite-type phases. Y_2BaF_8 is a stable line compound that exists beyond 46 mol% YF_3 in

* Corresponding author. Fax: +91-22-550-5151 or +91-22-550-9613.
E-mail address: aktyagi@magnum.barc.ernet.in (A.K. Tyagi).

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a BaF_2–YF_3 system. We have reported [1–3] several cubic ordered phases in SrF_2–YF_3 and CaF_2–YF_3 systems.

There are only a few reports on the MgF_2–YF_3 system. Those by Nafziger et al. [4] and Sobolev and coworkers [5] were restricted to high-temperature quenched samples. Sobolev and coworkers [5] mentioned the absence of any binary compounds in the MgF_2–YF_3 system. They found the eutectic composition for the MgF_2–YF_3 system at 36 mol% MgF_2 and 64 mol% YF_3, at the eutectic temperature of 974°C. They performed phase analysis on the quenched samples and reported that about 6 mol% MgF_2 could be retained in the cubic YF_3, in other words, the high-temperature modification. Trnovcova et al. [6] have reported the phase relation of the MgF_2–GdF_3 system. They studied the microstructure and ionic conductivity of these composites. They did not observe any binary compounds.

To investigate the room-temperature phase equilibria and probability of formation of any ordered phases in the MgF_2–YF_3 system, we have carried out a systematic study under slow-cooled conditions.

2. Experimental

About 25 compositions with the general formula $Y_{1-x}Mg_{2+1.5x}F_7$ ($-1.33 \leq x \leq 1$), starting from MgF_2 and YF_3, were prepared by the same procedure as followed in our earlier studies [1–3]. XRD patterns were recorded on a Philips PW1710 X-ray diffractometer with Ni-filtered Cu Kα radiation.

3. Results and discussion

To our surprise, we observed the lines β-YF_3 (orthorhombic) in the XRD patterns starting from the first composition, $Y_{0.1}Mg_{3.35}F_7$ (about 3 mol% YF_3 in the CaF_2 lattice). The other compositions showed an increase in intensity of YF_3 with concomitant reduction in that of MgF_2. A similar observation was reported by Sobolev and coworkers [5] in their samples quenched from 900°C. Earlier, Nafziger et al. [4] had reported the phase diagram for a MgF_2–YF_3 system above 1000°C.

The present study shows that no binary compound exists in the MgF_2–YF_3 system, throughout the composition range. The whole range of composition always ended with a mixture of MgF_2 and YF_3. We determined the lattice parameters of different phases present in some of the compositions. There was no change in lattice parameters as a function of x for MgF_2 or YF_3. Hence, it is inferred that there was no solubility of MgF_2 in YF_3 or of YF_3 in MgF_2. This behavior may be due to the large size difference between Mg^{2+} and Y^{3+} ions. It is not common for a large cation such as Y^{3+} to have an octahedral coordination in fluorides. Hence, there is no solubility of YF_3 in MgF_2. The rutile-type structure of MgF_2, in which Mg^{2+} is sixfold coordinated, could be the reason that MgF_2 has no solubility in orthorhombic YF_3. Earlier we observed [1–3] that CaF_2, SrF_2, and BaF_2 were also not soluble in β-YF_3.

Conclusions

The $Y_{1-x}Mg_{2+1.5x}F_7$ system does not show the formation of single-phase compounds. Under slow-cooled conditions, there is no solubility of YF_3 in the MgF_2 lattice or of MgF_2 in YF_3. In addition to ionic size, the role of the structure of the host lattice appears to play a role in determining the solubility of YF_3. However, it would be interesting to quench these samples from still higher temperatures or to record in situ high-temperature XRD patterns. Also, in order to convincingly correlate the role of ionic radius and structure of the host lattice in deciding solubility limits and existence of binary phases, it would be worthwhile to study the phase relations in MgF_2–ScF_3 and ZnF_2–YF_3 systems.

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